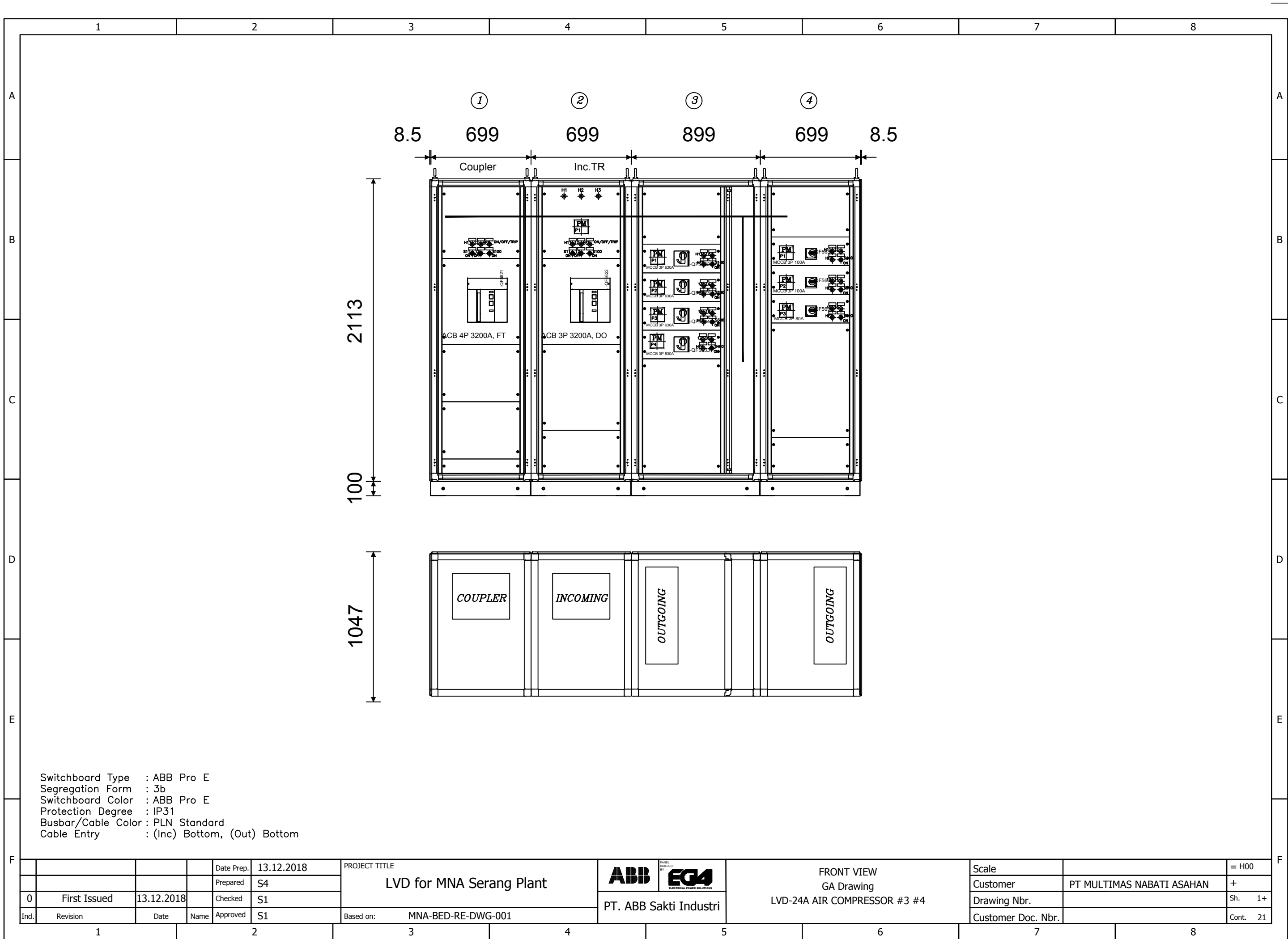


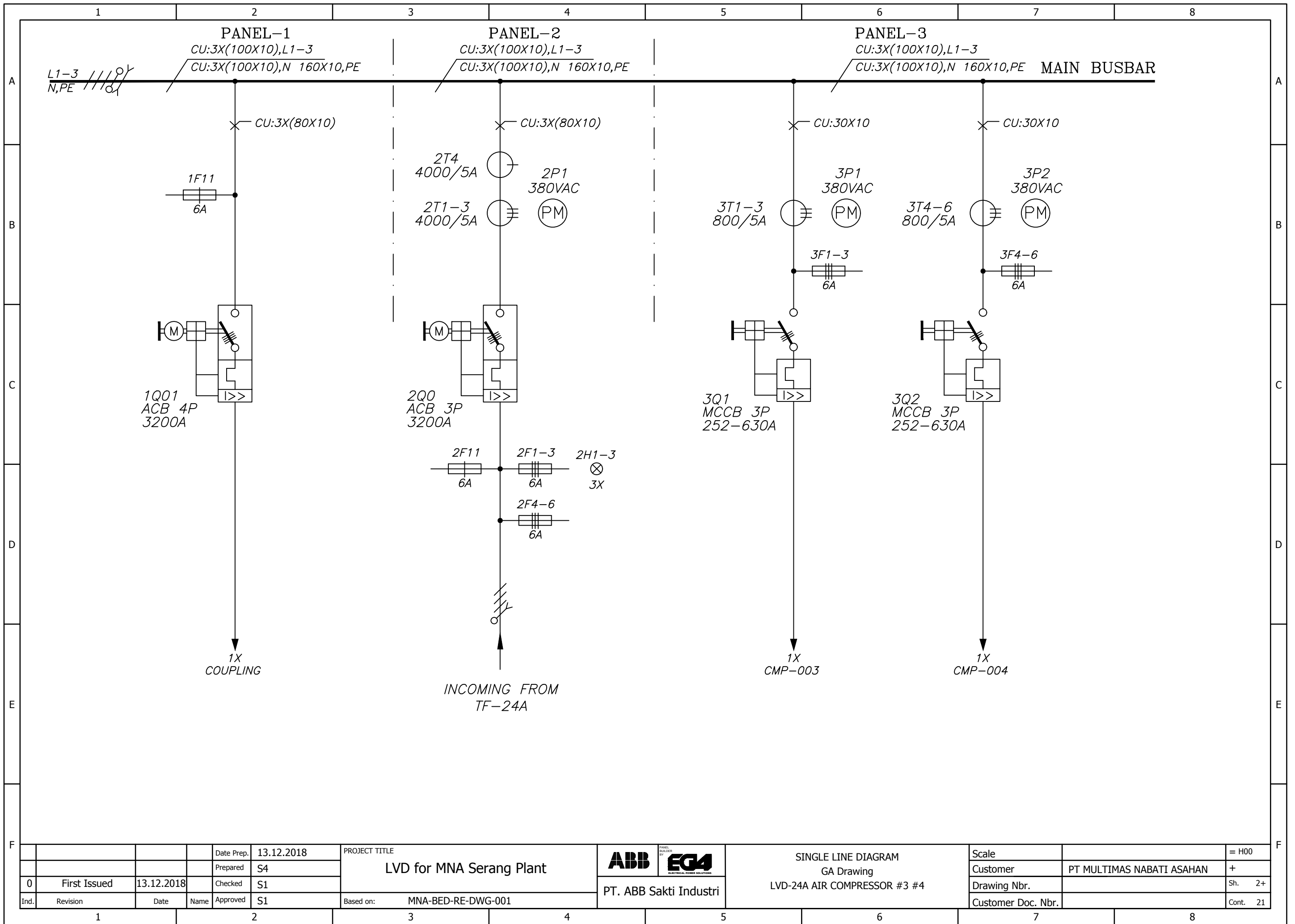
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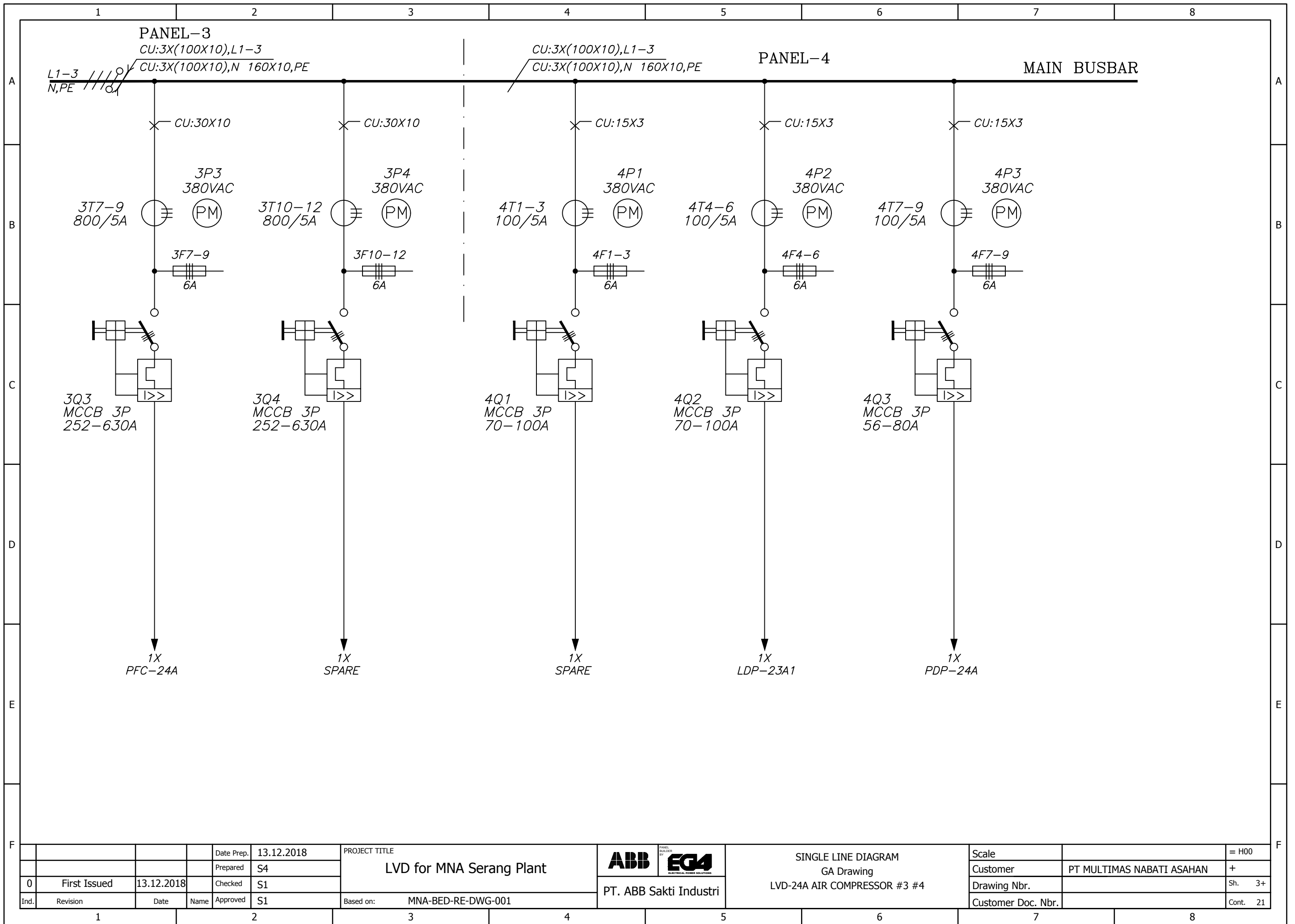
Switchboard Type : ABB Pro E  
Segregation Form : 3b  
Switchboard Color : ABB Pro E  
Protection Degree : IP31  
Busbar/Cable Color : PLN Standard  
Cable Entry : (Inc) Bottom, (Out) Bottom

|      |              |            |      |            |            |                                                  |                                                                                                                                                                                                       |                                                          |                    |                           |          |
|------|--------------|------------|------|------------|------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------|---------------------------|----------|
|      |              |            |      | Date Prep. | 13.12.2018 | PROJECT TITLE<br><b>LVD for MNA Serang Plant</b> |  <br>PT. ABB Sakti Industri | FRONT VIEW<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4 | Scale              |                           | = H00    |
|      |              |            |      | Prepared   | S4         |                                                  |                                                                                                                                                                                                       |                                                          | Customer           | PT MULTIMAS NABATI ASAHAN | +        |
| 0    | First Issued | 13.12.2018 |      | Checked    | S1         |                                                  |                                                                                                                                                                                                       |                                                          | Drawing Nbr.       |                           | Sh. 1+   |
| Ind. | Revision     | Date       | Name | Approved   | S1         | Based on:                                        | MNA-BED-RE-DWG-001                                                                                                                                                                                    |                                                          | Customer Doc. Nbr. |                           | Cont. 21 |

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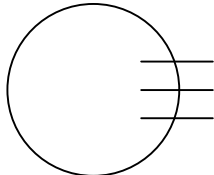
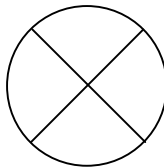
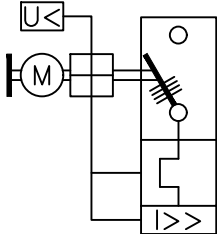
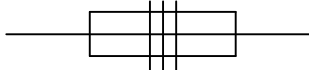
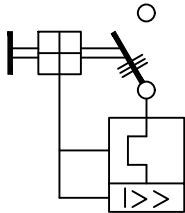


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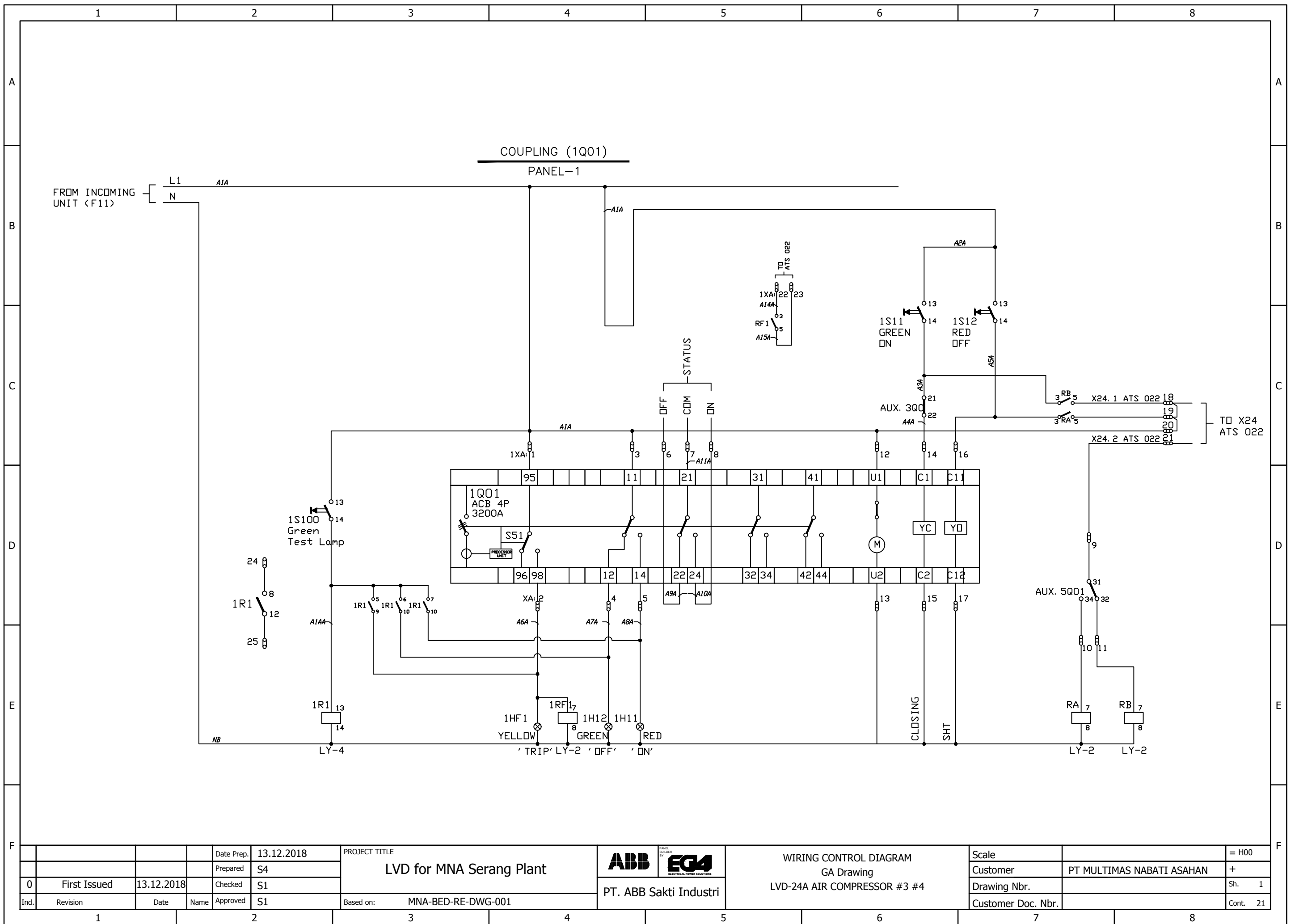


|      |          |              |            |                                           |    |                                          |  |                                                                   |  |                    |                           |          |
|------|----------|--------------|------------|-------------------------------------------|----|------------------------------------------|--|-------------------------------------------------------------------|--|--------------------|---------------------------|----------|
|      |          | Date Prep.   | 13.12.2018 | PROJECT TITLE<br>LVD for MNA Serang Plant |    | ABB<br>EG4<br>ELECTRICAL POWER SOLUTIONS |  | SINGLE LINE DIAGRAM<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4 |  | Scale              |                           | = H00    |
|      |          | Prepared     | S4         |                                           |    |                                          |  |                                                                   |  | Customer           | PT MULTIMAS NABATI ASAHAN | +        |
| 0    |          | First Issued | 13.12.2018 | Checked                                   | S1 |                                          |  |                                                                   |  | Drawing Nbr.       |                           | Sh. 3+   |
| Ind. | Revision | Date         | Name       | Approved                                  | S1 | Based on: MNA-BED-RE-DWG-001             |  | PT. ABB Sakti Industri                                            |  | Customer Doc. Nbr. |                           | Cont. 21 |

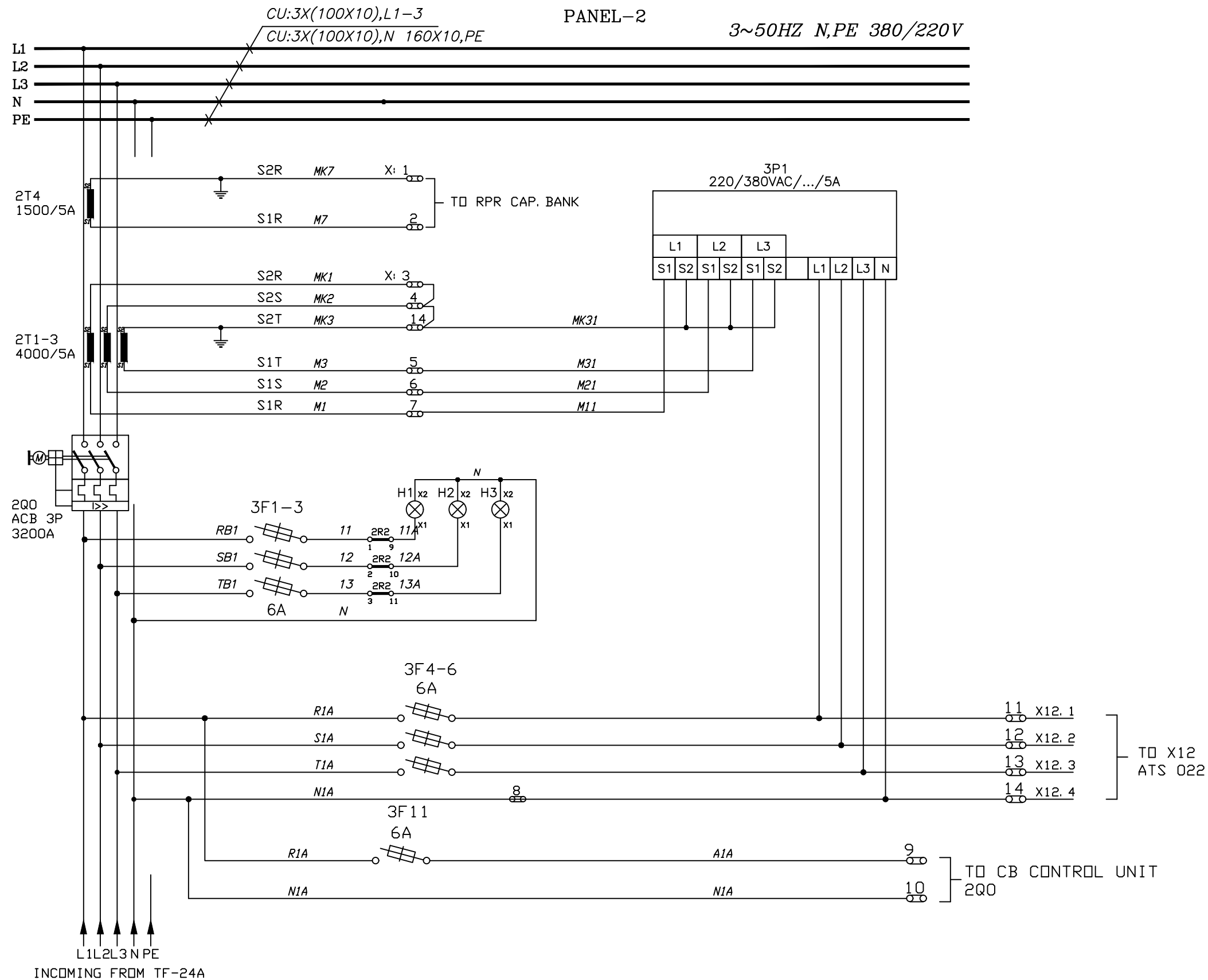
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|      |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    |   |  |
|------|--------------|-------------------------------------------------------------------------------------|------|---------------------|------------|-------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------|--|--------------------|---------------------------|-------|----|---|--|
| 1    |              | 2                                                                                   |      | 3                   |            | 4                                         |  | 5                                                                                                                                                                                                     |  | 6                                                                                   |  | 7                  |                           | 8     |    |   |  |
| A    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | A |  |
|      |              |    |      | CT                  |            |                                           |  |                                                                                                                                                                                                       |  |  |  | PILOT LAMP         |                           |       |    |   |  |
| B    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | B |  |
|      |              |    |      | ACB                 |            |                                           |  |                                                                                                                                                                                                       |  |  |  | FUSE               |                           |       |    |   |  |
| C    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | C |  |
|      |              |   |      | MCCB                |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    |   |  |
| D    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | D |  |
|      |              |  |      | DIGITAL POWER METER |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    |   |  |
| E    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | E |  |
| F    |              |                                                                                     |      |                     |            |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  |                    |                           |       |    | F |  |
|      |              |                                                                                     |      | Date Prep.          | 13.12.2018 | PROJECT TITLE<br>LVD for MNA Serang Plant |  |  <br>PT. ABB Sakti Industri |  | LEGEND<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4                                |  | Scale              |                           | = H00 |    |   |  |
|      |              |                                                                                     |      | Prepared            | S4         |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  | Customer           | PT MULTIMAS NABATI ASAHAN | +     |    |   |  |
| 0    | First Issued | 13.12.2018                                                                          |      | Checked             | S1         |                                           |  |                                                                                                                                                                                                       |  |                                                                                     |  | Drawing Nbr.       |                           | Sh.   | 4+ |   |  |
| Ind. | Revision     | Date                                                                                | Name | Approved            | S1         | Based on: MNA-BED-RE-DWG-001              |  |                                                                                                                                                                                                       |  |                                                                                     |  | Customer Doc. Nbr. |                           | Cont. | 21 |   |  |
| 1    |              | 2                                                                                   |      | 3                   |            | 4                                         |  | 5                                                                                                                                                                                                     |  | 6                                                                                   |  | 7                  |                           | 8     |    |   |  |

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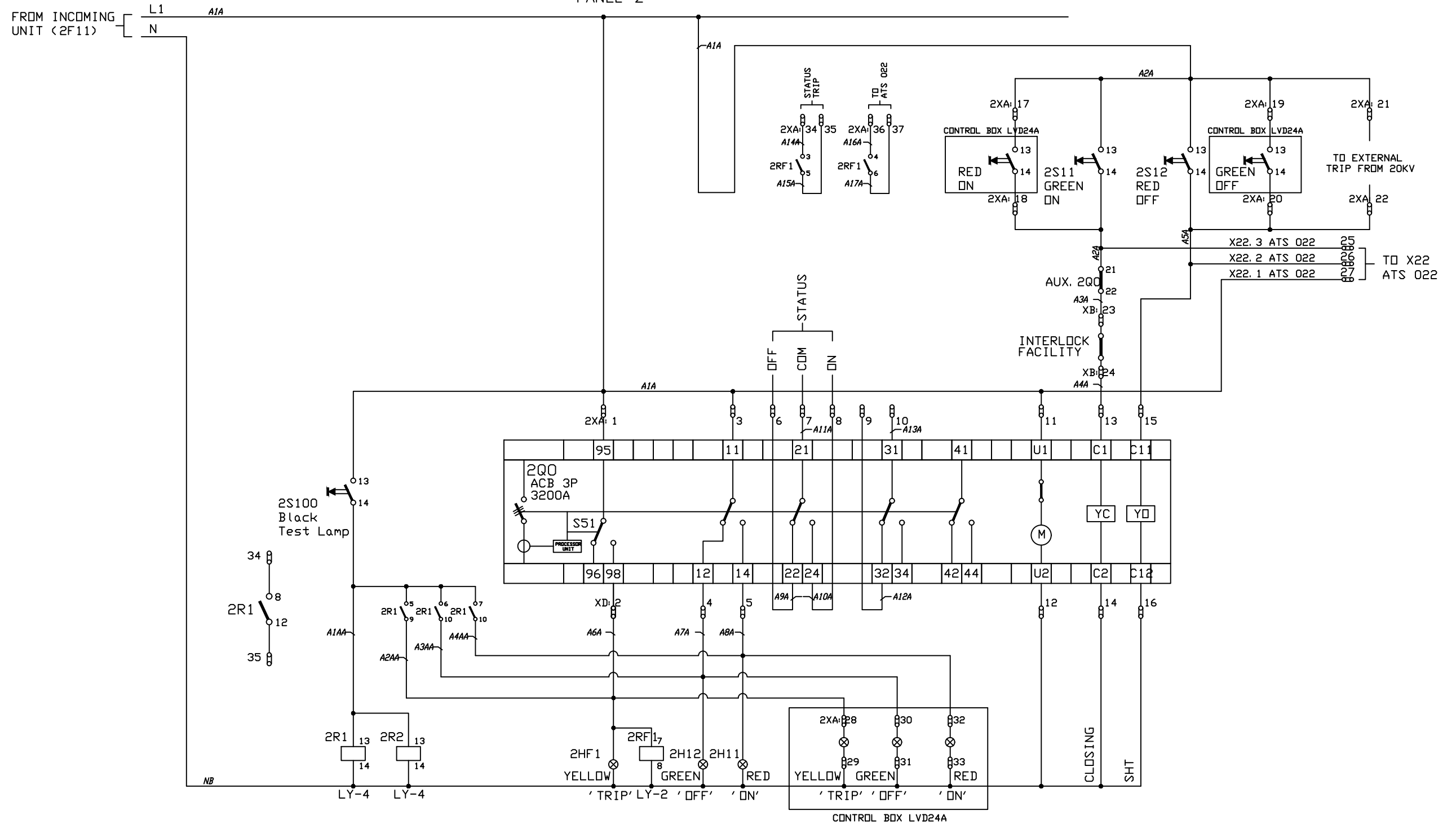


|      |              |            |            |            |                                                  |                                                                             |                                                                      |                    |                           |          |
|------|--------------|------------|------------|------------|--------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------|---------------------------|----------|
|      |              |            | Date Prep. | 13.12.2018 | PROJECT TITLE<br><b>LVD for MNA Serang Plant</b> | <b>ABB</b><br>PANEL BUILDING BY<br><b>EG4</b><br>ELECTRICAL POWER SOLUTIONS | WIRING CONTROL DIAGRAM<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4 | Scale              |                           | = H00    |
|      |              |            | Prepared   | S4         |                                                  |                                                                             |                                                                      | Customer           | PT MULTIMAS NABATI ASAHAN | +        |
| 0    | First Issued | 13.12.2018 | Checked    | S1         |                                                  |                                                                             |                                                                      | Drawing Nbr.       |                           | Sh. 1    |
| Ind. | Revision     | Date       | Name       | Approved   | S1                                               | Based on:                                                                   | MNA-BED-RE-DWG-001                                                   | Customer Doc. Nbr. |                           | Cont. 21 |



# CB CONTROL UNIT (2Q0)

PANEL-2

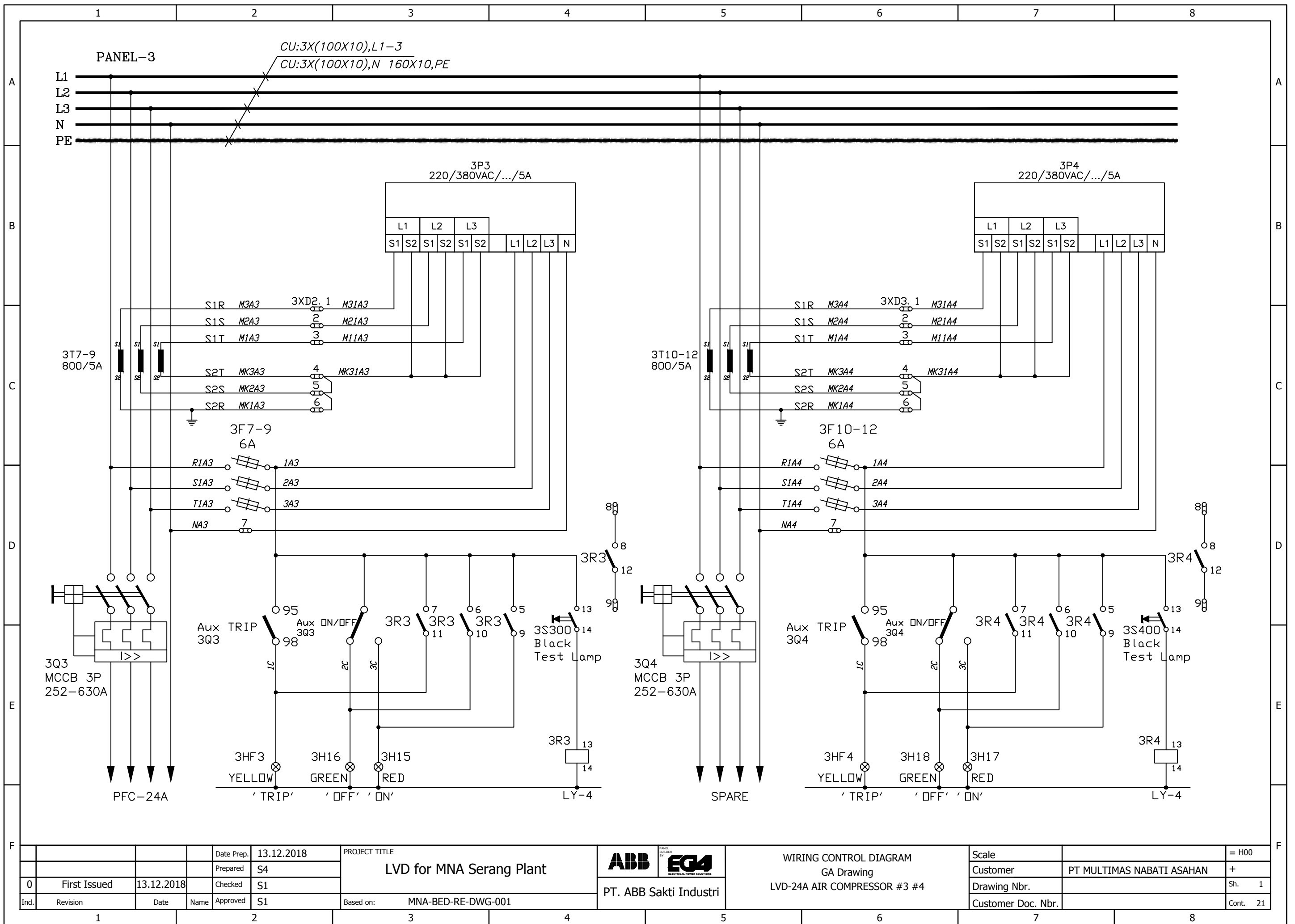


|      |          |              |            |                                           |    |                            |                                                                      |                    |                           |          |
|------|----------|--------------|------------|-------------------------------------------|----|----------------------------|----------------------------------------------------------------------|--------------------|---------------------------|----------|
|      |          | Date Prep.   | 13.12.2018 | PROJECT TITLE<br>LVD for MNA Serang Plant |    | <br>PT. ABB Sakti Industri | WIRING CONTROL DIAGRAM<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4 | Scale              |                           | = H00    |
|      |          | Prepared     | S4         |                                           |    |                            |                                                                      | Customer           | PT MULTIMAS NABATI ASAHAN | +        |
| 0    |          | First Issued | 13.12.2018 | Checked                                   | S1 |                            |                                                                      | Drawing Nbr.       |                           | Sh. 1    |
| Ind. | Revision | Date         | Name       | Approved                                  | S1 | Based on:                  | MNA-BED-RE-DWG-001                                                   | Customer Doc. Nbr. |                           | Cont. 21 |



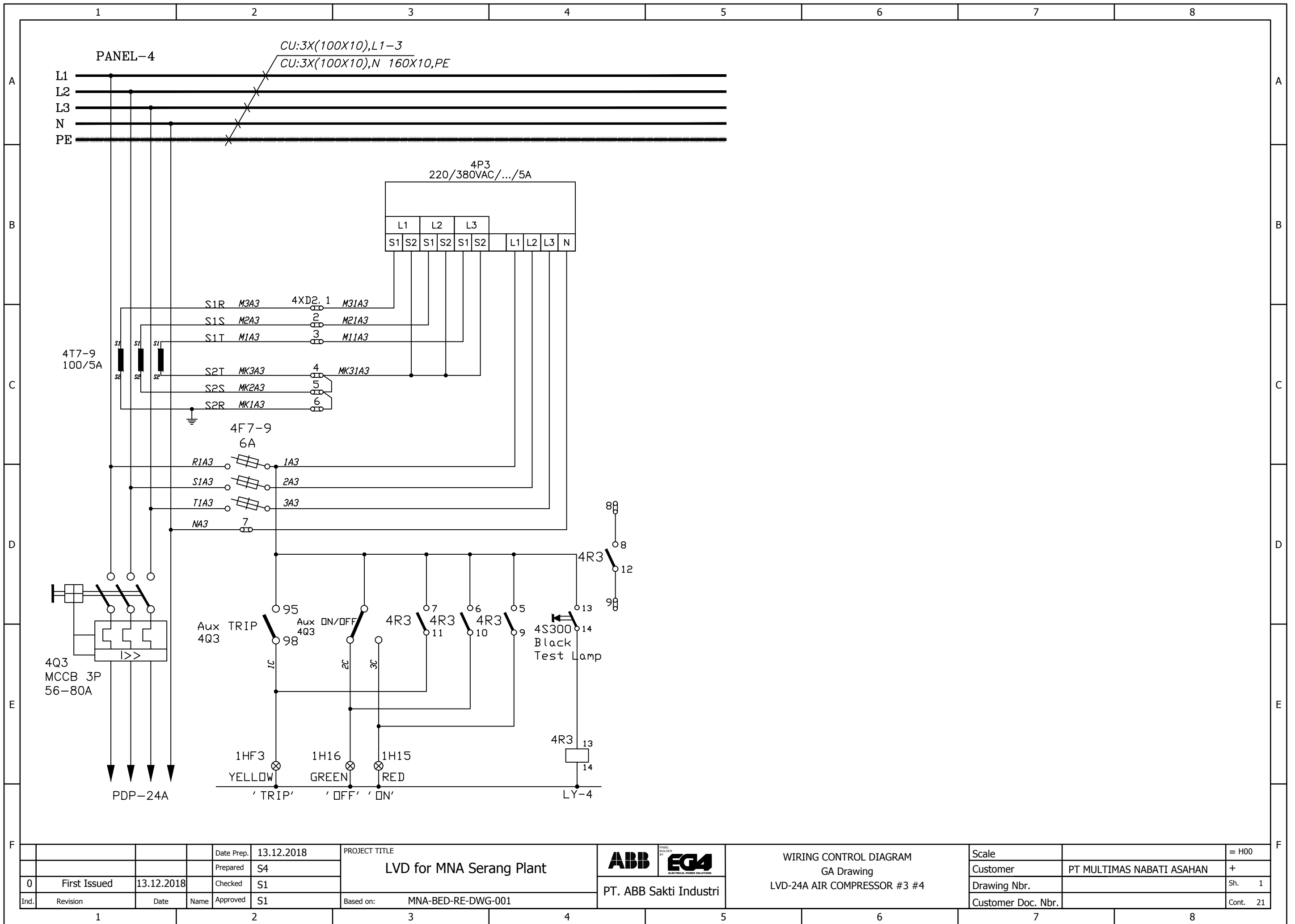


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|   |      |                                                      |            |      |                    |            |                                           |                    |                                                                                                                                                                                                          |                                                              |          |              |                           |        |   |
|---|------|------------------------------------------------------|------------|------|--------------------|------------|-------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------|--------------|---------------------------|--------|---|
| 1 |      | 2                                                    |            | 3    |                    | 4          |                                           | 5                  |                                                                                                                                                                                                          | 6                                                            |          | 7            |                           | 8      |   |
| A | N O  | DESCRIPTION                                          |            |      | TECHNICAL DATA     |            |                                           | QTY                | DESIGNATION                                                                                                                                                                                              |                                                              |          |              |                           |        |   |
|   |      | INCOMING                                             |            |      |                    |            |                                           |                    |                                                                                                                                                                                                          |                                                              |          |              |                           |        |   |
| B | 1    | Fuse Carrier 6A                                      |            |      | AXLE/Setara        |            |                                           | 7                  | 2F11,1F1-7                                                                                                                                                                                               |                                                              |          |              |                           |        |   |
|   | 2    | Pilot Lamp (LED) 220Vac/50Hz, Red                    |            |      | AXLE/Setara        |            |                                           | 2                  | 2H1,11                                                                                                                                                                                                   |                                                              |          |              |                           |        |   |
|   | 3    | Pilot Lamp (LED) 220Vac/50Hz, Yellow                 |            |      | AXLE/Setara        |            |                                           | 2                  | 2H2,HF1                                                                                                                                                                                                  |                                                              |          |              |                           |        |   |
|   | 4    | Pilot Lamp (LED) 220Vac/50Hz, Green                  |            |      | AXLE/Setara        |            |                                           | 2                  | 2H3,12                                                                                                                                                                                                   |                                                              |          |              |                           |        |   |
|   | 5    | DIGITAL POWER METER (*)                              |            |      | ABB, M2M ETHERNET  |            |                                           | 1                  | 2P1                                                                                                                                                                                                      |                                                              |          |              |                           |        |   |
|   | 6    | ATS 022 (*)                                          |            |      | ABB, ATS 022       |            |                                           | 1                  | -                                                                                                                                                                                                        |                                                              |          |              |                           |        |   |
| C | 7    | CT 4000/5A CL.0.5 40VA                               |            |      | Howig, HG 120-4000 |            |                                           | 4                  | 2T1-4                                                                                                                                                                                                    |                                                              |          |              |                           |        |   |
|   | 8    | ACB 3P 3200A, 66KA, DO, Ekip Touch LSIG (*)          |            |      | ABB, E4.2N32 D/O   |            |                                           | 1                  | 2Q0                                                                                                                                                                                                      |                                                              |          |              |                           |        |   |
|   | 9    | + Shunt Trip Release (Y0) (*)                        |            |      | ABB, Y0            |            |                                           | 1                  | -                                                                                                                                                                                                        |                                                              |          |              |                           |        |   |
|   | 10   | + Motor (M) + Closing Release (YC) For E2.2-E6.2 (*) |            |      | ABB, M+YC          |            |                                           | 1                  | -                                                                                                                                                                                                        |                                                              |          |              |                           |        |   |
|   | 11   | + Padlocks For E2.2-E6.2 (*)                         |            |      | ABB,               |            |                                           | 1                  | -                                                                                                                                                                                                        |                                                              |          |              |                           |        |   |
|   | 12   | + Ekip Com Protocol IEC 61850 (*)                    |            |      | ABB,               |            |                                           | 1                  | -                                                                                                                                                                                                        |                                                              |          |              |                           |        |   |
|   | 13   | Control Relay 220Vac                                 |            |      | Omron, LY-2        |            |                                           | 1                  | RF1                                                                                                                                                                                                      |                                                              |          |              |                           |        |   |
|   | 14   | Control Relay 220Vac                                 |            |      | Omron, LY-4        |            |                                           | 2                  | R1-2                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 15   | Push Button Test Lamp , 1NO, Black                   |            |      | Ega                |            |                                           | 1                  | S100                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 16   | Push Button On, 1NO, Green                           |            |      | Ega                |            |                                           | 1                  | 2S11                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
| D | 17   | Push Button Off, 1NO, Red                            |            |      | Ega                |            |                                           | 1                  | 2S12                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 18   |                                                      |            |      |                    |            |                                           |                    |                                                                                                                                                                                                          |                                                              |          |              |                           |        |   |
|   | 19   | COUPLER                                              |            |      |                    |            |                                           |                    |                                                                                                                                                                                                          |                                                              |          |              |                           |        |   |
|   | 20   | Fuse Carrier 6A                                      |            |      | AXLE/Setara        |            |                                           | 1                  | 1F8                                                                                                                                                                                                      |                                                              |          |              |                           |        |   |
|   | 21   | Pilot Lamp (LED) 220Vac/50Hz, Red                    |            |      | AXLE/Setara        |            |                                           | 1                  | 1H11                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 22   | Pilot Lamp (LED) 220Vac/50Hz, Yellow                 |            |      | AXLE/Setara        |            |                                           | 1                  | 1HF1                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
| E | 23   | Pilot Lamp (LED) 220Vac/50Hz, Green                  |            |      | AXLE/Setara        |            |                                           | 1                  | 1H12                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 24   | ACB 4P 3200A, 66KA, FT, Ekip Dip LI (*)              |            |      | ABB, E4.2N32 F/T   |            |                                           | 1                  | 1Q01                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 25   | Push Button On/Test Lamp , 1NO, Black                |            |      | Ega                |            |                                           | 1                  | S100                                                                                                                                                                                                     |                                                              |          |              |                           |        |   |
|   | 26   | Control Relay 220Vac                                 |            |      | Omron, LY-4        |            |                                           | 1                  | R1                                                                                                                                                                                                       |                                                              |          |              |                           |        |   |
| F |      |                                                      |            |      | Date Prep.         | 13.12.2018 | PROJECT TITLE<br>LVD for MNA Serang Plant |                    | <br><br>PT. ABB Sakti Industri | COMPONENT LIST<br>GA Drawing<br>LVD-24A AIR COMPRESSOR #3 #4 |          | Scale        |                           | = H00  |   |
|   |      |                                                      |            |      | Prepared           | S4         |                                           |                    |                                                                                                                                                                                                          |                                                              |          | Customer     | PT MULTIMAS NABATI ASAHAN | +      |   |
|   | 0    | First Issued                                         | 13.12.2018 |      | Checked            | S1         |                                           |                    |                                                                                                                                                                                                          |                                                              |          | Drawing Nbr. |                           | Sh. A+ |   |
|   | Ind. | Revision                                             | Date       | Name | Approved           | S1         | Based on:                                 | MNA-BED-RE-DWG-001 | Customer Doc. Nbr.                                                                                                                                                                                       |                                                              | Cont. 21 |              |                           |        |   |
|   | 1    |                                                      | 2          |      | 3                  |            | 4                                         |                    | 5                                                                                                                                                                                                        |                                                              | 6        |              | 7                         |        | 8 |

